

Margo Kersey

UCSF Dyslexia Center
Department of Neurology
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EDUCATION

2016-2020 University of California, Los Angeles
B.S. Applied Mathematics, Minor in Cognitive Science
GPA: 3.702/4.00

Notable Coursework: Machine Learning, Mathematical Modeling, Probability, Networks, Optimization, Python with Applications, Philosophy of Mind, Cognitive Development

Undergraduate Projects:

- COVID-19 Classification using Convolutional Neural Networks with Chest X-rays
- A Centrality-based Analysis of the Bitcoin Transaction Network

GRANTS, HONORS, & SCHOLARSHIPS

2024 UCSF WINGS Employee Development Grants for School of Medicine Career Staff Employees
2019 German Academic Exchange Service (DAAD) Scholarship for Research Internships in Science and Engineering
2018 UCLA Psychology Undergraduate Research Grant
2016-2020 Dean's List, UCLA College of Letters & Science

RESEARCH EXPERIENCE

Research Data Analyst

UCSF Dyslexia Center, Department of Neurology
San Francisco, CA

2022-Present

- Create machine learning pipelines and Python modules to predict neurodevelopmental diagnoses and cognitive profiles from MRI and multidisciplinary behavioral evaluations
- Harmonize and process multimodal MRI data (structural, diffusion, functional) for 500+ research subjects using BIDS pipelines and custom-built post-processing scripts
- Implement custom diffusion tractography, integrate neuropsychological and neuroimaging measures, and perform analyses for lab projects and grant applications
- Automate lab workflows via Python scripting, including database-wide interactive visualization platforms, subject-level diffusion MRI presentations, REDCap API-powered database visualizations

- Co-lead UCSF Memory and Aging Center Artificial Intelligence Office Hours to teach staff, postdocs, and faculty how to implement AI tools in their research

Research Assistant

Forschungszentrum Jülich, Institute of Neuroscience and Medicine

Jülich, Germany

2019

- Conducted correlation analyses between resting-state fMRI network measures and psychological metrics (resilience and personality traits) for manuscript published in *Nature Scientific Reports* (Altinok et al., 2021)
- Fine-tuned and implemented custom EEG artifact removal pipeline in MATLAB, improving data quality of auditory mismatch negativity study comparing healthy and schizophrenic populations
- Assisted with EEG setup for trimodal brain imaging data collection (MR-PET-EEG)
- Provided editing and grammar review for 3 scientific manuscripts prepared by international research team

Research Assistant

Computational Vision and Learning Lab, UCLA Department of Psychology

Los Angeles, CA

2018-2019

- Applied deep convolutional neural networks (AlexNet, VGG, ResNet, DenseNet) using Python's Keras and TensorFlow for research investigating computational models of human vision, contributing to lab research published in *PLOS Computational Biology*
- Administered and conducted preliminary analysis of MATLAB experimental tasks studying temporal boundaries in human memory, supporting research published in the *Journal of Vision*

PUBLICATIONS

1. Altinok, D. C. A., Rajkumar, R., Nießen, D., Sbaihat, H., **Kersey, M.**, Shah, N. J., Veselinović, T., & Neuner, I. (2021). Common neurobiological correlates of resilience and personality traits within the triple resting-state brain networks assessed by 7-Tesla ultra-high field MRI. *Scientific reports*, 11(1), 11564.
<https://doi.org/10.1038/s41598-021-91056-y>

MANUSCRIPTS IN PREPARATION

1. **Kersey, M.** *et al.* (in preparation). Characterizing Neural Signatures of Dyslexia and Co-occurring Math Learning Difficulties (MLD) with Machine Learning.
2. **Kersey, M.** *et al.* (in preparation). Data-driven Cognitive and Neuroimaging Clusters in Persistent Dyslexia.

3. Palser, E., **Kersey, M.** *et al.* (in preparation). Language outcomes and structural anatomical correlates of early language delay in children with dyslexia during middle childhood.
4. Rauschecker, A., Carpenter, E., **Kersey, M.** *et al.* (in preparation). Imaging Individual Differences at the Intersection of ADHD and Dyslexia: Gray Matter and White Matter Microstructure.

POSTERS & PRESENTATIONS

1. **Kersey, M.** *et al.* (2024). *Data-driven Cognitive and Neuroimaging Clusters in Persistent Dyslexia*. Flux Society Annual Meeting, Baltimore, MD, United States. <https://tinyurl.com/fluxsociety2024>
2. Carpenter, E., **Kersey, M.** *et al.* (2024) *Examining an Adaptive Assessment of Executive Functions (ACE-X) in Children with Dyslexia*. Flux Society Annual Meeting, Baltimore, MD, United States. <https://tinyurl.com/fluxsociety2024>
3. **Kersey, M.** *et al.* (2024) *Characterizing Neural Signatures of Dyslexia and Co-occurring Math Learning Difficulties (MLD) with Machine Learning*. Cognitive Neuroscience Society Annual Meeting, Toronto, ON, Canada. <https://tinyurl.com/cogneurosociety2024>
4. Martin-Moreno, D.V., ..., **Kersey, M.** *et al.* (2025). *Validity of the Adaptive Cognitive Evaluation Explorer (ACE-X) as a Battery of Executive Functioning in Children with Developmental Dyslexia*. Poster accepted for presentation at the International Neuropsychological Society Annual Meeting, New Orleans, LA, United States.

COMMUNITY ENGAGEMENT

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| 2023-Present | Organizing Committee & Event Staff, UCSF Brain Health Events
Brain health education, resources, and activities provided to BIPOC communities |
| 2023-Present | Founder, New Wave Clay
Handmade ceramics sold locally in San Francisco, CA |
| 2023 | Organizing Committee, Bay Area Brainhack
Joint UCSF-UC Berkeley-Stanford neuroscience hackathon open to the Bay Area community as part of Brainhack Global |
| 2023 | Attendee, NeuroHackademy, University of Washington eScience Institute
Intensive summer school in neuroimaging and data science |
| 2017-2020 | Founder and President, Irish Dancers at UCLA
On-campus student organization celebrating Irish culture through dance |

TECHNICAL SKILLS

- Programming: Python (Scikit-learn, Nilearn, pyAFQ, DIPY, Antspyx, Nibabel, SciPy, Statsmodels, Stepmix, PCNtoolkit, Brainstat, LangChain, Pandas, NumPy, Keras, Tensorflow, Joblib, IPywidgets, Regex, Matplotlib, Seaborn, Plotly, Voila, Streamlit), Jupyter, MATLAB, Bash, Singularity, Docker, Github, Remote Servers & HPC
- Neuroimaging: sMRIPrep, QSIprep, fMRIPrep, FreeSurfer, FreeView, MRICroGL, ITK-Snap, MI-Brain
- Machine Learning & AI: Neural Networks, Regression, SVM, Feature Selection, Clustering, LLMs (OpenAI, Huggingface, Mistral, Ollama)
- Other Software: Microsoft Office Suite (Outlook, Word, PowerPoint, Excel), Adobe Illustrator

REFERENCES

Maria Luisa Gorno Tempini

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Hongjing Lu

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